

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 12th February, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 1	CLOs to be covered: CLO1
Lab Title: Input/Output, Data Types, Operators, Conditional Statements (If/Else)	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1: Input/Output, Data Types, Operators, Conditional Statements (If/Else)

Lab Goals/Objectives:

To understand basic input/output operations, Python data types, operators, and to apply conditional statements (if/else) to control program flow.

Equipment Required:

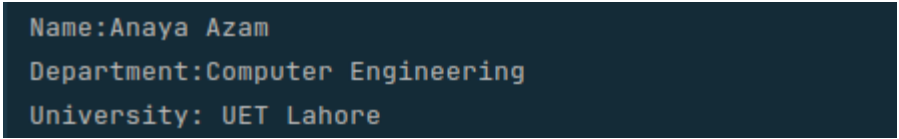
Python 3, VS Code / PyCharm

Theory:

Python provides built-in functions such as `print()` and `input()` for interacting with the user. Every value in Python has a data type (`int`, `float`, `bool`, `str`, etc.), and operators allow us to perform arithmetic, comparison and logical operations on these values. Conditional statements (`if/elif/else`) allow a program to make decisions and execute different blocks of code based on given conditions.

Lab Work:**Task 1: Print Your Information**

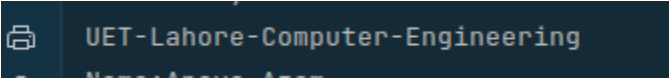
```
print("Name:Anaya Azam")
print("Department:Computer Engineering")
print("University: UET Lahore")
```

Result:

```
Name:Anaya Azam
Department:Computer Engineering
University: UET Lahore
```

Task 2: Using sep and end

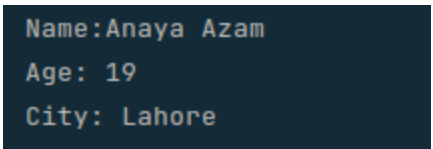
```
print("UET","Lahore","Computer","Engineering",sep="-")
```

Result:

```
UET-Lahore-Computer-Engineering
```

Task 3: Format Output

```
print("Name:Anaya Azam\nAge: 19\nCity: Lahore")
```

Result:

```
Name:Anaya Azam
Age: 19
City: Lahore
```

Task 4: Take Name and Age

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print("Name:", name)
print("Age:", age)
```

Result:

```
Enter your name: anaya
Enter your age: 19
Name: anaya
Age: 19
```

Task 5: Identify Data Types

```
x=10
y=3.14
z=True
name="UET"
print(x)
print(y)
print(z)

print(type(x))
print(type(y))
print(type(z))
print(type(name))

print(len(name))
```

Result:

```
10
3.14
True
<class 'int'>
<class 'float'>
<class 'bool'>
<class 'str'>
3
```

Task 6: Percentage (Conditions, Grades)

```
students = int(input("Enter number of students: "))
for i in range(students):
    name = input("Enter name: ")
    roll_no = int(input("Enter roll number: "))
    obt_marks = int(input("Enter your marks: "))
```

```
total_marks = int(input("Enter your total marks: "))
print(name)
print(roll_no)
print(obt_marks)
print(total_marks)
if obt_marks < total_marks:
    percentage = (obt_marks / total_marks)* 100
    print (percentage)
    if percentage >= 90:
        print("A")
    elif percentage >= 85:
        print("A-")
    elif percentage >= 80:
        print("B+")
    elif percentage >= 75:
        print("B")
    elif percentage >= 70:
        print("B-")
    elif percentage >= 65:
        print("C+")
    elif percentage >= 60:
        print("C")
    elif percentage >= 55:
        print("C-")
    elif percentage >= 50:
        print("D")
    elif percentage < 50:
        print("F")
elif obt_marks > total_marks:
    print ("Not Valid")
elif total_marks == 0:
    print ("Not Defined")
elif obt_marks == total_marks:
    print ("100%")
```

Result:

```
Enter name: anaya
Enter roll number: 12
Enter your marks: 280
Enter your total marks: 300
anaya
12
280
300
93.33333333333333
A
```

Task 7: String Concatenation

```
x="25"  
y="anaya"  
print(x)  
print(y)  
print(x+y)
```

Result:



```
25  
anaya  
25anaya
```

Conclusions:

Through these tasks, basic I/O operations, Python data types, operators, and conditional (if/else) logic were practiced and understood.